Yao Ji

H. Milton Stewart Postdoctoral Fellow

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Postdoc

Georgia Institute of Technology, Atlanta, Georgia Oct. 2024–Present
H. Milton Stewart Postdoctoral Fellow
H. Milton Stewart School of Industrial and Systems Engineering (ISyE)
Mentor: Guanghui (George) Lan
Current focus: Parameter-free stochastic optimization; variational inequalities; gradient norm minimization; high-order optimization; reinforcement learning; reinforcement learning in healthcare.

Education

Purdue University, West Lafayette, Indiana Aug. 2019–Aug. 2024
Ph.D. in Operations Research
Committee: Harsha Honnappa (advisor), Raghu Pasupathy, Alex Wang, and Gesualdo Scutari (advisor)
Dissertation: *High-Dimensional Inference over Networks: Statistical and Computational Guarantees*

Beijing Normal University, Beijing, China Aug. 2016–Jun. 2019
M.S. in Probability and Mathematical Statistics
advisor: Wenming Hong
Thesis: *Conditional Limit Theorem for Bellman–Harris Branching Process*

Beijing Normal University, Beijing, China Aug. 2012–Jun. 2016
B.S. in Mathematics
Thesis: *Conceptual New Proofs of Geometric Convergence of Moment Generating Function for Galton–Watson Process in the Noncritical Case*

Research Interests

Stochastic optimization; reinforcement learning; healthcare.

Research Papers

Journal Papers

Yao Ji, Guanghui Lan, and Jason Zhu
Computation of Strong Solutions to Stochastic Variational Inequalities
Journal manuscript | Under review, *Mathematical Programming* — first-round review; arXiv:2609.04188 (2026).

Yao Ji and Guanghui Lan
Stochastic Auto-conditioned Fast Gradient Methods with Optimal Rates
Job-market paper | Journal manuscript | Under review, *Mathematical Programming* — first-round review; arXiv:2604.06525 (2026).

Yao Ji and Guanghui Lan
High-order Accumulative Regularization for Gradient Minimization in Convex Programming
Journal manuscript | R&R, *SIAM Journal on Optimization* — second-round review; arXiv:2511.03723 (2025).

Yao Ji, Gesualdo Scutari, Ying Sun, and Harsha Honnappa
Distributed Sparse Regression via Penalization
Journal article | Published, *Journal of Machine Learning Research*, 24(272):1–62, 2023. Paper.

Yao Ji, Gesualdo Scutari, Ying Sun, and Harsha Honnappa
Distributed (ATC) Gradient Descent for High Dimension Sparse Regression
Journal article | Published, *IEEE Transactions on Information Theory*, 69(8):5253–5276, 2023. Paper.

Vladimir A. Vatutin, Wenming Hong, and **Yao Ji**
Reduced critical Bellman–Harris branching processes for small populations
Journal article | Published, *Discrete Mathematics and Applications*, 28(5):319–330, 2018. Paper.

Conference and Workshop Papers

Ruqi Bai, **Yao Ji**, Mingyu Kim, Easton Currie, Zeyu Zhou, and David I. Inouye
Provable Robustness to Spurious Correlations via Invariant Data for Robust Finetuning
Workshop paper | Accepted, AISTATS 2026 Workshop on Causality in the Age of AI Scaling. Paper.

Ruqi Bai, **Yao Ji**, Zeyu Zhou, and David I. Inouye
From Invariant Representations to Invariant Data: Provable Robustness to Spurious Correlations via Noisy Counterfactual Matching
Preprint | arXiv:2505.24843 (2025).

Work in Progress

Yao Ji, Yan Li, Guanghui Lan, Zehua Dong, Shuoling Li, Ilker Guven, Xiaoyu Chen, Lihui Bai, and Jiapeng Huang
Transformer-Based Reinforcement Learning for Perioperative Management of Cardiac Surgery-Associated Acute Kidney Injury

Ruqi Bai*, **Yao Ji***, and Guanghui Lan
Benchmarking Stochastic Auto-conditioned Fast Gradient Methods
*Equal contribution.

Yao Ji and Guanghui Lan
A Unified Stochastic Auto-conditioned Single-loop Method for Gradient Norm Minimization with Optimal Rates

Teaching Experience

Georgia Institute of Technology, Stewart School of Industrial and Systems Engineering
Advisor
ISyE 4106: Senior Design Fall 2026

Purdue University, Industrial Engineering
Teaching Assistant
IE 335: Operations Research Spring and Fall 2023
IE 330: Probability and Statistics in Engineering Fall 2022
IE 590: Introduction to Optimization Algorithms (graduate-level) Fall 2022

Beijing Normal University, School of Mathematical Sciences
Co-Lecturer
Large Deviation Theory Spring 2018
Brownian Motion Fall 2017
Random Walk in Random Environment Spring 2017
Galton–Watson Branching Process Fall 2016

Teaching Assistant; Outstanding Teaching Assistant Award recipient
Measure Theory I Fall 2017 and Fall 2018
Measure Theory II Spring 2017 and Spring 2018
Stochastic Calculus for Finance (graduate-level) Fall 2016

Outreach

Instructor: Pre-College Reinforcement Learning Course for High School Students, Georgia Tech Summer 2026
Judge: STEM Spark: INFORMS K–12 Poster Competition 2026
Judge: Georgia Tech 2026 CRIDC Poster Competition 2026

Awards and Honors

H. Milton Stewart Postdoctoral Fellowship, Georgia Institute of Technology 2024
Lee A. Chaden Fellow IE Scholarship, Purdue University 2023
Graduate School Summer Research Grant, Purdue University 2023
Travel Grant, School of Industrial Engineering, Purdue University 2022, 2023
Ross Fellowship, Purdue University 2019, 2020
Dr. Theodore J. and Isabel M. Williams Fellowship in Industrial Control Systems, Purdue University 2019
First Prize Scholarship (2nd of 53, School of Mathematics), Beijing Normal University 2018
First Prize Scholarship (1st of 12, Markov Process), Beijing Normal University 2017
Outstanding Teaching Assistant for Measure Theory, Beijing Normal University 2017
Outstanding Undergraduate Thesis, School of Mathematics, Beijing Normal University 2016
First Prize Scholarship (top 5%), School of Mathematics, Beijing Normal University 2015
Second Prize Scholarship, Beijing Normal University 2014
Second Prize, China Undergraduate Mathematical Modeling Contest (top 5%) 2014

Talks

Stochastic Auto-conditioned Fast Gradient Methods with Optimal Rates
- MOPTA, Lehigh University, Bethlehem, PA Aug. 2026
- Clemson University, School of Mathematical and Statistical Sciences, Clemson, SC Apr. 2026
- INFORMS Optimization Society Conference, Atlanta, GA Mar. 2026

High-order Accumulative Regularization for Gradient Minimization in Convex Programming
- INFORMS Annual Meeting, Atlanta, GA Oct. 2025

Distributed (ATC) Gradient Descent for High Dimension Sparse Regression
- International Symposium on Mathematical Programming (ISMP), Montréal, Canada Jul. 2024

Poster Presentations

Stochastic Auto-conditioned Fast Gradient Methods with Optimal Rates
- Southern California OR/OM Day, USC Marshall, Los Angeles, CA May 2026

Transformer-Based Patient Response Modeling Across All Perioperative Phases for Cardiac Surgery-Associated Acute Kidney Injury Management
- Society of Cardiovascular Anesthesiologists Annual Meeting, Nashville, TN Apr. 2026

Distributed (ATC) Gradient Descent for High Dimension Sparse Regression
- SIAM Conference on Mathematics of Data Science, Atlanta, GA Oct. 2024
- Midwest Machine Learning Symposium, University of Illinois Chicago, Chicago, IL May 2023
- Statistics and Optimization in Data Science Workshop, Purdue University, West Lafayette, IN May 2023

Distributed Sparse Regression via Penalization
- Cornell ORIE Young Researchers Workshop, Ithaca, NY Oct. 2023
- International Conference on Continuous Optimization (ICCOPT)/MOPTA, Lehigh University, Bethlehem, PA Jul. 2022

Service

Journal Reviewer: Mathematical Programming; *SIAM Journal on Optimization*; *Operations Research*; *Journal of the American Statistical Association*; *Stochastic Systems*; *Journal of Global Optimization*; *Journal of Optimization Theory and Applications*; and *IEEE Transactions on Automatic Control*.

Conference Reviewer: IEEE International Symposium on Information Theory.

References

Guanghui (George) Lan
(A. Russell Chandler III Chair and Professor)
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Arkadi Nemirovski
(John P. Hunter, Jr. Chair and Professor)
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Jiapeng Huang, M.D., Ph.D.
(Enterprise Chair of Anesthesiology and the Wasmuth Endowed Chair in Anesthesiology Research)
Cleveland Clinic
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David I. Inouye
(Associate Professor of Electrical and Computer Engineering)
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